

Use It or Lose It?

Programmable money, negative rates and the design of the digital euro

Zheng Huang · Topics of Central Banking · October 2026

Speaker script · 20 minutes + 10 minutes Q&A

THE STORY, SLIDE BY SLIDE

- 2 This talk follows one question from stimulus vouchers to the design of the digital euro.
- 3 When the central bank creates money that nobody spends, monetary policy loses its grip.
- 4 Interest rates could not be cut far below zero, because cash always pays zero.
- 5 Vouchers with an expiry date get money spent, which is exactly what a liquidity trap needs.
- 6 A programmable digital euro could build this pressure into money itself, in one of two ways.
- 7 To compare the two designs, consider a holder who must search for good uses of money.
- 8 An expiry date pushes out more spending, but it ends in a fire sale and splits money into vintages.
- 9 Continuous decay keeps every euro equal, and it is nothing more than a negative interest rate.
- 10 Yet the draft regulation fixes the digital euro's interest rate at exactly zero.
- 11 A zero-rate digital euro is cash without storage costs, so it pushes the lower bound back up to zero.
- 12 The lower bound and disintermediation are the same problem: a central-bank rival to bank deposits.
- 13 A hard holding limit stops a run in a crisis, but it also blocks ordinary users in normal times.
- 14 When costs have a cliff and benefits do not, price and quantity together beat either one alone.
- 15 The optimal digital euro has three numbers: a free tier, a Pigouvian rate above it, and a cap.
- 16 The hybrid is the only design that works in normal times, at negative rates and in a crisis.
- 17 The cap should move with the cycle: higher when banks hold ample liquidity, lower when buffers are thin.
- 18 The digital euro should bear interest: zero for payments, decay above, and a cap that moves with the cycle.

TIMING

1	0:15	10	0:40
2	0:45	11	1:10
3	1:15	12	1:00
4	1:15	13	1:20
5	1:15	14	1:30
6	0:50	15	1:30
7	1:15	16	1:00
8	1:40	17	0:50
9	1:15	18	0:50

Checkpoints: end of slide 6 at 5:35; slide 9 at 9:45; slide 13 at 13:55; slide 18 at 19:35, leaving 25 seconds of slack.

NUMBERS TO KNOW

- Eurosystem balance sheet: €2.2tn (2014) to €8.6tn (2021); inflation below 2% in 2014–20.
- Deposit facility rate ≤ 0 from June 2014 to July 2022; lowest -0.50% (2019–22).
- China coupons: ¥3 per ¥1. France: one-month MPC 61% vs 23%.
- Expiry vs decay, 30 days: 84% vs 73% useful spending; 16% dumped; value 0.80 to 0.50.
- Decay of 1% a year raises velocity by about 12%.
- Art. 16(8): “shall not bear interest”. Cap range €1,500–3,000.
- Crisis outflow: €699bn at €3,000 (ECB); model €648bn; no cap €4.7tn.
- Design: free tier €510, decay -2.4% ($2.65\% - 0.26\%$), cap €3,000.
- Aggregate LCR 157%; cap €1,300 at 130%, €5,600 at 185%.

SLIDE 1

0:15

Use It or Lose It?

Programmable money, negative rates and the design of the digital euro

Zheng Huang
Topics of Central Banking
October 2026

SAY

- “Good morning. My talk is about what happens if a central bank programs a holding cost into money, and what that means for the design of the digital euro.”

SLIDE 2

0:45

OVERVIEW

This talk follows one question from stimulus vouchers to the design of the digital euro

Should the digital euro bear interest, and if so, in what form?

- THE PROBLEM
Monetary policy stalls when new money is hoarded and interest rates cannot fall below the cash floor.
- A TEMPTING FIX
Vouchers with an expiry date get money spent, and a programmable digital euro could carry one.
- A BETTER FIX
An expiry date ends in a fire sale and splits money into vintages; continuous decay is simply a negative rate.
- THE TWIST
The draft regulation fixes the rate at zero, which makes the digital euro costless cash and raises the lower bound.
- THE DESIGN
Combining price and quantity gives a free tier, a Pigouvian rate above it, and a cap that moves with the cycle.

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SAY

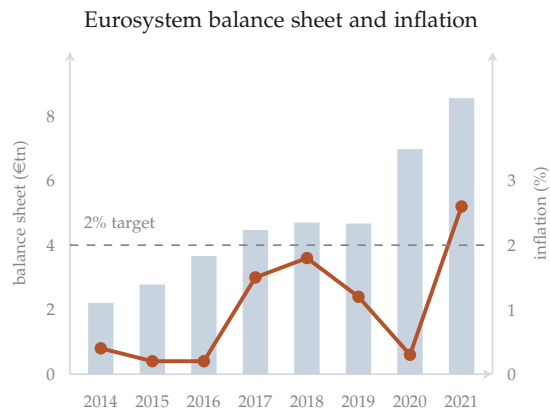
- “The whole talk answers one question: should the digital euro bear interest, and in what form?”
- Walk down the five steps in one sentence each; tell the audience the answer comes in three numbers at the end.

SLIDE 3

1:15

THE PROBLEM · 1

When the central bank creates money that nobody spends, monetary policy loses its grip



Sources: ECB annual consolidated balance sheet of the Eurosystem (end of year); Eurostat HICP, annual average.

Between 2014 and 2021 the Eurosystem almost quadrupled its balance sheet, from €2.2 trillion to €8.6 trillion.

Inflation nevertheless stayed below the 2% target in every year from 2014 to 2020, and was close to zero in 2014–16 and 2020.

In the quantity equation $MV = PY$, the extra money M was offset by a fall in velocity V : households and firms held the new money instead of spending it.

This is the first blockage: money is created, but it is not spent.

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SAY

- The bars are the Eurosystem balance sheet: almost four times larger in seven years.
- The line is inflation: below target every year until 2021.
- In $MV = PY$, more M did not raise PY because V fell. People held the money.
- “That is the first blockage: money is created but not spent.”

IF ASKED

Q Wasn't the balance sheet growth meant to lower long-term rates, not to raise spending directly?

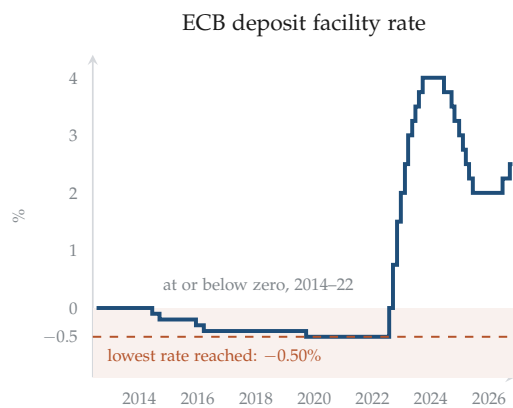
A Yes, asset purchases worked through yields and portfolio rebalancing. The point is only that the extra base money did not translate into proportional spending.

SLIDE 4

1:15

THE PROBLEM · 2

Interest rates could not be cut far below zero, because cash always pays zero



Source: ECB key interest rates.

The ECB cut its deposit facility rate below zero in June 2014 and kept it at or below zero until July 2022, but it never went lower than -0.50% .

The reason is the outside option of cash. Banknotes pay zero, so if banks charged savers more than the cost of storing and insuring cash, savers would withdraw their money.

The lower bound is therefore not exactly zero but minus the cost of holding cash: $i \geq -c$.

This is the second blockage: interest rates cannot fall far enough to make people spend.

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SAY

- The deposit facility rate went below zero in June 2014 and stayed at or below zero for eight years.
- It stopped at -0.50% . Below that, savers would pull money into banknotes, which pay zero.
- The true floor is minus the cost of holding cash: storage, insurance, transport.
- “That is the second blockage: rates cannot fall far enough.”

IF ASKED

Q Inflation is now above target. Why does this matter?

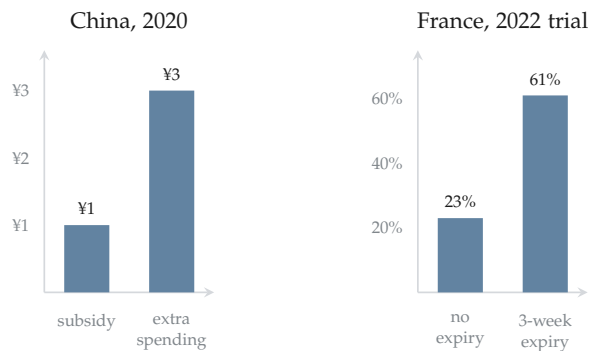
A The floor bound for most of 2009–2021, and the digital-euro regulation will last decades. It should work in both regimes.

SLIDE 5

1:15

A TEMPTING FIX

Vouchers with an expiry date get money spent, which is exactly what a liquidity trap needs



In 2020 Chinese cities distributed digital coupons through mobile payment platforms that had to be used within about a week. Each yuan of subsidy generated roughly three yuan of additional spending.

In a randomised experiment in France, households received prepaid cards. When the balance expired after three weeks, 61% was spent within a month, against 23% without an expiry date.

A deadline turns money that would have been saved into spending today.

Xing, Zou, Yin, Wang & Li (AEJ: Macro 2023); Ding, Jiang, Msall & Notowidigdo (2024); Boehm, Fize & Jaravel (2024), one-month MPC.

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SAY

- Governments found a way around the first blockage: vouchers with a deadline.
- Left: Chinese digital coupons, roughly three yuan of extra spending for each yuan of subsidy.
- Right: a French randomised trial; the expiry date nearly triples the one-month spending rate.

IF ASKED

Q The Chinese coupons also had minimum-spend thresholds. Isn't that what drives the effect?

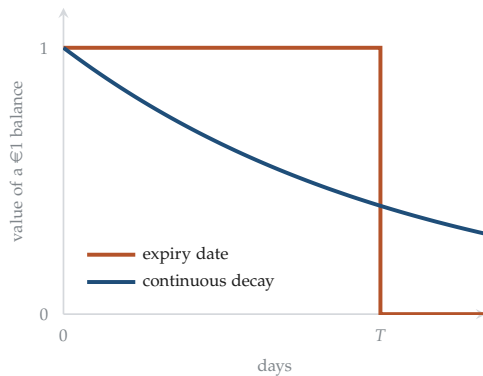
A Partly, as Ding et al. show. That is why I add the French trial, which isolates the deadline.

SLIDE 6

0:50

A TEMPTING FIX

A programmable digital euro could build this pressure into money itself, in one of two ways



A central bank digital currency is programmable, so the central bank could charge holders for keeping money idle.

With an **expiry date**, nothing happens until day T , and then the whole balance is lost.

With **continuous decay**, also called demurrage, the balance shrinks by a small fraction every day, like Silvio Gesell's stamped money of the 1930s.

Both designs charge for holding money; they differ only in *when* the charge falls. **Which one should a central bank choose?**

SAY

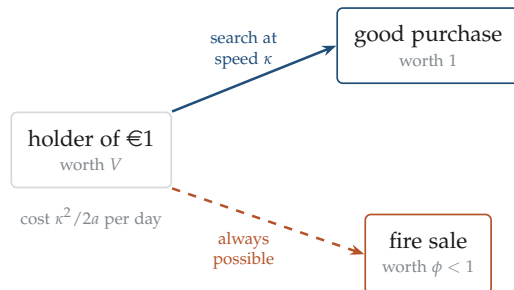
- A digital euro is programmable, so a holding cost could be written into money itself.
- Red: an expiry date; the balance is untouched until T , then gone. Blue: continuous decay, Gesell's idea; a little every day.
- "Same total cost, different timing. Which one should a central bank choose?"

SLIDE 7

1:15

THE MODEL

To compare the two designs, consider a holder who must search for good uses of money



Each euro is worth V to its holder. Finding a good use for it takes effort: searching at speed κ costs $\kappa^2/2a$ per day, and κ is the velocity of money. A bad use, such as buying something of little value or selling the token at a discount, is always available and worth $\phi < 1$.

Decay at rate λ : $(r + \lambda)V = \frac{a}{2}(1 - V)^2$

Expiry at T : $\frac{dV}{d\tau} = -rV + \frac{a}{2}(1 - V)^2$, $V = \phi$ at T

In both cases the holder spends at speed $\kappa^* = a(1 - V)$. The less money is worth holding, the faster it moves.

SAY

- To answer that, I need a simple model of how holders spend.
- A holder searches for a good use of each euro; faster search costs more. A fire sale is always possible but worth less.
- The value of holding money satisfies one equation under decay and a differential equation under expiry.
- “In both cases, the less money is worth holding, the faster it is spent.”

IF ASKED

Q Why a quadratic search cost?

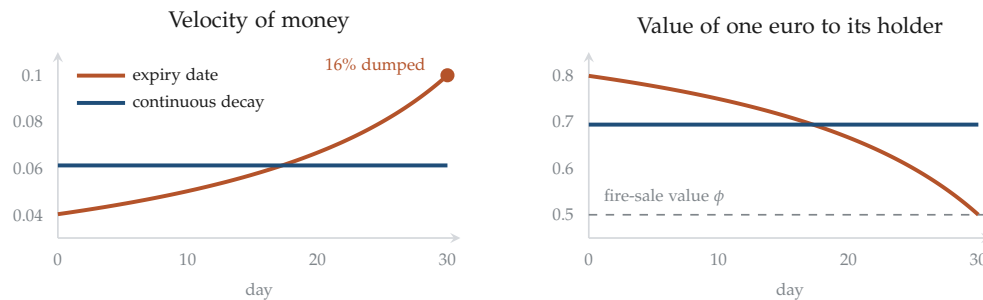
A It is the simplest convex cost with an interior solution. With a linear cost, velocity would literally explode at the deadline; convex is the conservative case.

SLIDE 8

1:40

RESULT 1

An expiry date pushes out more spending, but it ends in a fire sale and splits money into vintages



Both designs are calibrated to the same average velocity over 30 days. The expiry date gets more useful spending done, **84%** of issuance against **73%**, because it forces the whole balance out. But velocity accelerates into the deadline, and **16%** of issuance is still unspent on the last day and is dumped on goods worth half its value. Worse, an expiring euro is worth **0.80** when new and **0.50** just before it expires: one euro is no longer equal to another.

Parameters: $a = 0.2$, $\phi = 0.5$, $r = 4\%$ a year. Results at other horizons and parameters in the backup.

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SAY

- Both designs are tuned to the same average velocity over 30 days, so the race is fair.
- Left: velocity under expiry accelerates into the deadline; under decay it is flat.
- Expiry gets more useful spending done, **84%** against **73%**, because it forces the whole balance out. That surprised me.
- But **16%** is still unspent on the last day and goes to a fire sale.
- Right: an expiring euro is worth **0.80** when new and **0.50** when old. One euro is no longer equal to another.

IF ASKED

Q Why match average velocity rather than total spending?

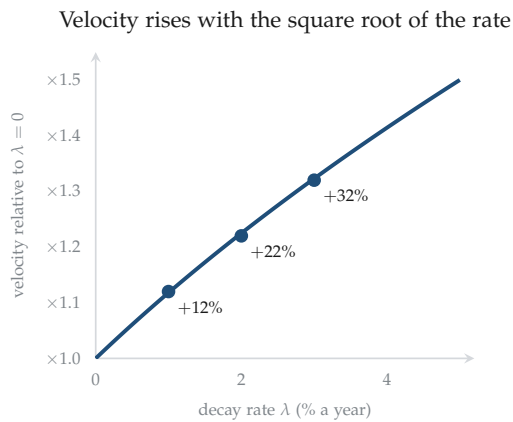
A Velocity is what the central bank wants to raise in a liquidity trap. Matching total spending is impossible, because decay always takes part of the balance.

SLIDE 9

1:15

RESULT 2

Continuous decay keeps every euro equal, and it is nothing more than a negative interest rate



Decay avoids every problem of the expiry date. There is no deadline and no fire sale, every euro has the same value whatever its age, and it can apply to all money rather than to a separate voucher.

A decay rate λ is simply an interest rate of $-\lambda$.

Velocity follows $\kappa^* \approx \sqrt{2a(r + \lambda)}$, the same square-root rule as in the Baumol–Tobin model of money demand: a decay of 1% a year raises velocity by about 12%.

Decay addresses both blockages: money moves faster, and a negative rate finally reaches retail balances. An expiry date belongs to a voucher; decay can belong to money.

SAY

- Decay avoids all three problems: no deadline, no fire sale, every euro worth the same.
- And decay at λ is just an interest rate of $-\lambda$.
- The curve: velocity rises with the square root of the rate, as in Baumol–Tobin. One percent of decay, 12% more velocity.
- “Decay clears both blockages: money moves faster, and a negative rate reaches retail balances.”

IF ASKED

Q Couldn't a smart contract simply track the age of each expiring euro?

A Yes, but then shops must accept old and new euros either at par, which invites arbitrage, or at different prices, which ends the singleness of money.

SLIDE 10

0:40

THE TWIST

Yet the draft regulation fixes the digital euro's interest rate at exactly zero

"The digital euro shall not bear interest."

European Commission, proposal for a regulation on the digital euro, Article 16(8), June 2023



The rate is not positive and not negative, in any circumstances. Holdings are also capped per person, with a range of €1,500–3,000 under discussion. The final text is being negotiated now, so the design question is still open.

SAY

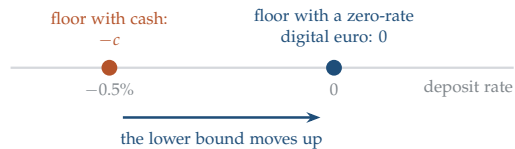
- “So the natural instrument is a rate on the digital euro. But the draft regulation rules it out.”
- Read the quote; point to the timeline. The text is in trilogue now, so the question is open.

SLIDE 11

1:10

THE TWIST

A zero-rate digital euro is cash without storage costs, so it pushes the lower bound back up to zero



A bank cannot pay depositors less than their best alternative in central-bank money, net of the cost of holding it:

$$r_D \geq \max_j \{ i_j - c_j \}.$$

Cash pays zero but costs something to store and insure, so the floor lies slightly below zero. This is why the ECB could reach -0.50% .

A zero-rate digital euro pays the same zero but costs nothing to hold. Up to the holding limit, it lifts the floor for deposit rates back to exactly zero, and takes away the little room below zero that the ECB had.

SAY

- A bank cannot pay less than the best central-bank alternative, net of the cost of holding it.
- Cash pays zero but is costly to hold, so the floor sits at $-c$. That is why -0.50% was possible.
- A zero-rate digital euro pays zero and costs nothing. Up to the cap, the floor moves back to zero.

IF ASKED

Q Retail deposit rates never went negative anyway.

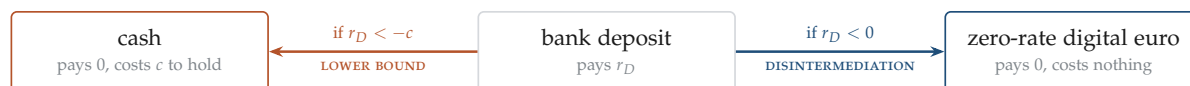
A For small balances banks chose not to charge. Large and corporate deposits did pay negative rates. A zero-rate digital euro turns zero into a legal floor for everyone, up to the cap.

SLIDE 12

1:00

THE TWIST

The lower bound and disintermediation are the same problem: a central-bank rival to bank deposits



The lower bound and bank disintermediation are usually discussed as separate issues. They are the same mechanism: whenever central-bank money offers a better net return than a bank deposit, depositors move.

A zero-rate digital euro therefore puts the most pressure on bank deposits precisely when rates are negative, which is when banks' margins are already squeezed. Fixing its rate at zero ties the digital euro's financial-stability risk to the state of monetary policy.

SAY

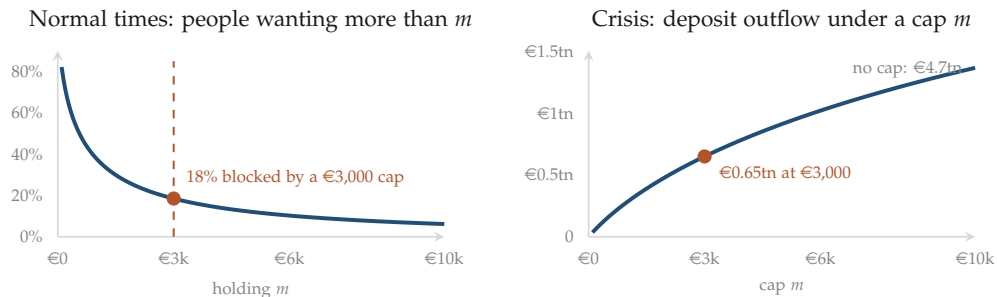
- Left arrow: deposits flow into cash when $r_D < -c$. That is the lower bound.
- Right arrow: deposits flow into the digital euro when $r_D < 0$. That is disintermediation.
- “One mechanism, two names. And with a zero rate, the risk peaks exactly when rates are negative.”

SLIDE 13

1:20

THE TRADE-OFF

A hard holding limit stops a run in a crisis, but it also blocks ordinary users in normal times



In normal times people hold the digital euro to make payments. In the ECB's survey, 18% would like to hold more than €3,000; a hard cap simply blocks them, while a price would let them hold more if they pay for it. In a crisis people run for safety, and a small interest-rate difference cannot stop them. **Price is enough in normal times; only quantity works in a crisis.**

Survey distribution: Lambert et al. (ECB WP 2980, 2024). Outflows: own calibration to ECB technical data on holding limits (Oct 2025), which reports €699bn at €3,000.

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SAY

- The ECB's answer is a hard cap, around €3,000.
- Left: in normal times, 18% of people would like to hold more; the cap blocks them.
- Right: in a crisis the cap is what matters. €0.65 trillion at €3,000 against €4.7 trillion with no cap.
- "A price is enough in normal times. In a crisis only a quantity works."

IF ASKED

Q Where does the crisis curve come from?

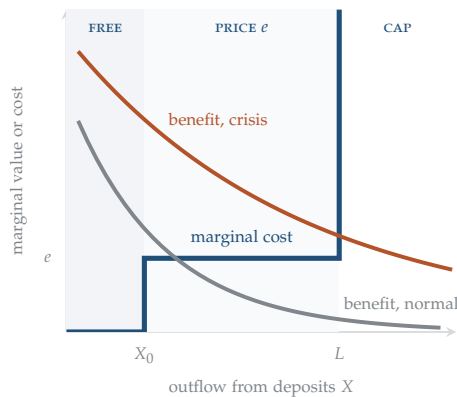
A The ECB published €156bn at a €500 cap and €699bn at €3,000. I fit the survey distribution of deposits to those two points. The implied population is 360 million, the actual euro-area population, and the no-cap outflow of €4.7tn is close to Meller and Soons' €5tn.

SLIDE 14

1:30

THE PRINCIPLE

When costs have a cliff and benefits do not, price and quantity together beat either one alone



Weitzman (1974) showed that under uncertainty a price works better when the benefit curve is steeper than the cost curve, and a quantity works better when the cost curve is steeper:

$$\Delta_{\text{price-quantity}} = \frac{\sigma^2 (b - c)}{2b^2}.$$

The marginal cost of deposit outflows has three segments: zero while banks can absorb them, a constant e while banks replace deposits with dearer funding, and vertical once their liquidity buffers run out.

In normal times demand crosses the flat part, where a price works best. In a crisis it shifts out and hits the cliff, where only a quantity works. **Both happen on the same curve, so the best instrument is a tiered price with a cap** (Roberts and Spence 1976).

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SAY

- Prices or quantities is an old question. Weitzman: a price is better when benefits are steeper than costs, a quantity when costs are steeper.
- The blue staircase is the cost of deposit outflows: free, then a constant e , then a cliff.
- Grey, normal times: demand crosses the flat part, so a price works. Red, crisis: demand hits the cliff, so only a cap works.
- “Both happen on one curve, so the answer is both: a tiered price with a cap.”

IF ASKED

Q Hasn't Bindseil already proposed tiering?

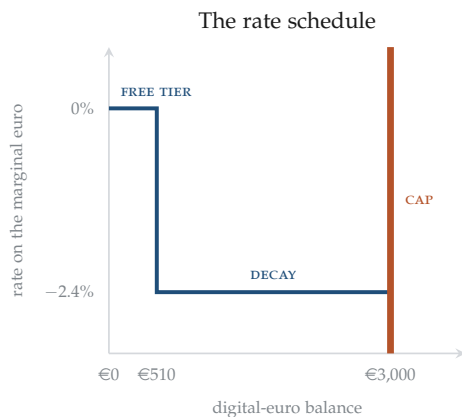
A Yes. What I add is the reason, from the shape of the cost curve, and a calibration of each tier.

SLIDE 15

1:30

THE DESIGN

The optimal digital euro has three numbers: a free tier, a Pigouvian rate above it, and a cap



Each segment of the cost curve gives one number.

Free tier: €510. Payments should only use the deposit outflow that banks absorb at no cost. The ECB expects payment digitalisation to bring €127bn of new deposits, and $N \mathbb{E}[\min(t, M_1)] = €127\text{bn}$ gives $M_1 \approx €510$.

Decay above the tier: -2.4%. Holders pay the bank's cost of replacing a deposit, the refinancing rate minus the deposit rate: $2.65\% - 0.26\%$.

Cap: €3,000. Crisis outflows stay within what banks can bear: $N \mathbb{E}[\min(d, \bar{M})] = \theta D$.

t : desired holding; d : sight deposits per person; $N = 360\text{m}$; θD : tolerated crisis outflow. Data: ECB WP 2980; ECB holding-limit analysis (Oct 2025); ECB rates and MIR statistics (2026).

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SAY

- Each segment of the staircase gives one number.
- Free tier €510: payments use only what banks absorb at no cost, the €127bn the ECB expects from digitalisation.
- Decay of 2.4% above it: the bank's cost of replacing a deposit, refinancing rate minus deposit rate.
- Cap €3,000: crisis outflows stay within what banks can bear.

IF ASKED

Q Isn't -2.4% a very large penalty?

A Almost nobody pays it; it only keeps the digital euro from becoming a savings account. It is also an upper bound: with central-bank pass-through funding the social cost is lower.

Q Why €127bn?

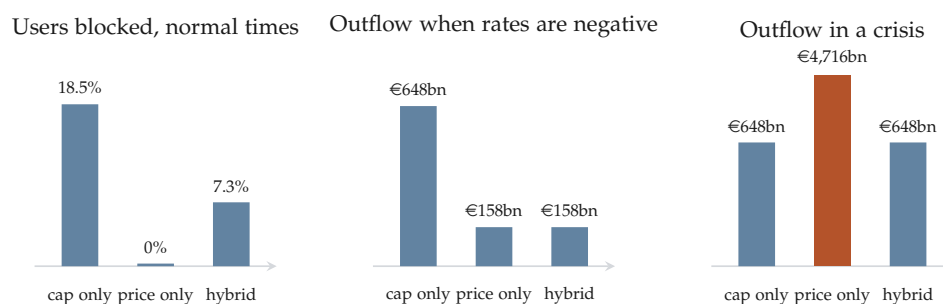
A It leaves banks no worse off than today. With €100–300bn the tier ranges from €370 to €1,840 (backup).

SLIDE 16

1:00

THE DESIGN

The hybrid is the only design that works in normal times, at negative rates and in a crisis



A cap alone blocks almost one user in five and still lets €648bn leave banks whenever deposit rates turn negative, because every capped euro is a zero-rate euro. A price alone handles normal times and negative rates, but in a crisis it would let €4.7 trillion run. **The hybrid keeps the cap's protection in a crisis, the price's protection at negative rates, and blocks fewer users**, because the decay above the tier shrinks demand for large balances.

The price-only crisis bar is truncated. Model calibrated to ECB survey and holding-limit data; see backup.

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SAY

- Three situations, three designs.
- A cap alone blocks almost one user in five and still leaks €648bn when rates are negative, because every capped euro pays zero.
- A price alone fails in a crisis.
- “Only the hybrid works in all three.”

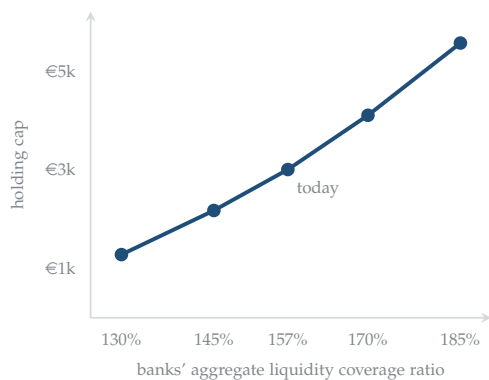
SLIDE 17

0:50

THE DESIGN

The cap should move with the cycle: higher when banks hold ample liquidity, lower when buffers are thin

A countercyclical cap



Aggregate LCR of significant institutions: 156.7% (ECB, Q3 2025). Scaling rule is illustrative.

The cap protects banks' liquidity, so it should depend on how much liquidity banks hold.

If the tolerated crisis outflow is proportional to banks' liquidity surplus above the 100% minimum, the cap becomes a function of the aggregate liquidity coverage ratio. At today's 157% it is €3,000. It would fall to about €1,300 at 130% and rise to about €5,600 at 185%.

The rule should be published and mechanical, so that a lower cap is anticipated rather than read as a signal of panic.

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SAY

- The cap protects bank liquidity, so it should depend on bank liquidity.
- If the tolerated outflow scales with the liquidity surplus, the cap goes from about €1,300 to €5,600 across plausible ratios.
- The rule must be published and mechanical, so that a cut is expected rather than read as panic.

IF ASKED

Q Wouldn't lowering the cap in a downturn trigger the run it is meant to prevent?

A That is why it must follow a published rule tied to published data, like a countercyclical capital buffer, not a discretionary announcement.

SLIDE 18

0:50

CONCLUSION

The digital euro should bear interest: zero for payments, decay above, and a cap that moves with the cycle

1. Deadlines get money spent, but an expiry date ends in a fire sale and makes one euro worth more than another. It belongs to a voucher, not to money.
2. Continuous decay keeps every euro equal and is simply a negative interest rate. It is the only form in which money itself can carry a holding cost.
3. A digital euro fixed at zero is cash without storage costs. It raises the lower bound and creates the most pressure on bank deposits when rates are negative.
4. The best design combines price and quantity: a free tier of about €500 for payments, decay of about 2.4% above it, and a cap tied to banks' liquidity.

Article 16(8) should be replaced by a rule that lets the digital euro bear interest in this form.

SAY

- Read the four points.
- Close: "Deadlines belong to vouchers; money can only carry a rate. Article 16(8) should be replaced by a rule that lets the digital euro bear interest: zero for payments, decay above, and a cap that moves with the cycle. Thank you."

MORE QUESTIONS

Q *Isn't negative interest on CBDC already in Bordo and Levin (2017)?*

A Yes, as an idea. The new parts here are why it must be decay rather than expiry, and how a rate and a cap should be combined and calibrated.

Q *Would people accept a digital euro that can shrink?*

A Balances up to the free tier never shrink. Above it, the charge only makes the digital euro no better than a bank account.

Q *The ECB says the digital euro will not be programmable money.*

A That means no restrictions on what it can buy. Decay restricts nothing; it is a rate on a balance.

Q *Your model is partial equilibrium.*

A Yes. The qualitative results do not depend on general equilibrium, and the magnitudes are calibrated to the ECB's own numbers.

Q *Do coupons just pull spending forward?*

A The evidence is mixed: Xing et al. (2023) find purchases moved forward, Ding et al. (2024) do not. The argument does not rely on it.

GLOSSARY

Velocity	Spending per unit of money per period, $V = PY/M$.
MPC	Share of an extra euro spent within a period.
Lower bound (ZLB / ELB)	The rate below which savers switch to cash; slightly below zero because cash is costly to hold.
Demurrage / decay	A continuous charge on holding money; a negative interest rate.
Singleness of money	Every form of a currency exchanges one-for-one at par.
Baumol–Tobin	Inventory model of money demand; holdings fall with the square root of the holding cost.
Disintermediation	Deposits leaving banks for central-bank money.
Weitzman (1974)	Prices beat quantities when benefits are steeper than costs, and vice versa.
Roberts–Spence (1976)	A price schedule with a quantity ceiling beats either instrument alone.
Pigouvian rate	A charge equal to the cost one holder imposes on others.
LCR	Liquidity coverage ratio: liquid assets over 30-day stressed outflows; minimum 100%.
MRO	Main refinancing operations rate, 2.65% since 16 Sep 2026.
Trilogue	Negotiation between Commission, Council and Parliament on EU legislation.
Art. 16(8)	Draft regulation: "The digital euro shall not bear interest."